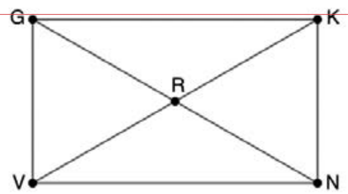
Geometry
Unit 8
Lesson 3 Homework

Name Answer Key_____________
Date____________________
Period___________________

Rectangle $GKNV$ is shown with diagonals $\overline{GN}$ and $\overline{KV}$ intersecting at R where $GK = 26x - 13$ and $NV = 14x + 5$.



1. Select all true properties of $GKNV$.
   - ☐ Opposite sides are parallel.
   - ☐ Opposite sides are congruent.
   - ☐ Consecutive angles are congruent.
   - ☐ All angles are right angles.
   - ☐ Diagonals are perpendicular.
   - ☐ All sides are congruent.

2. What is the value of x?

   $x = 1.5$

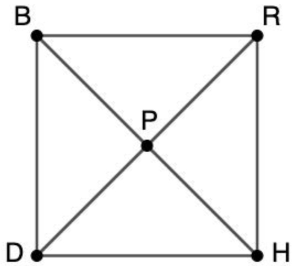
3. What is the length of $\overline{NV}$?

   $NV = 26$

Square $BRHD$ with diagonals $\overline{BH}$ and $\overline{RD}$ intersecting at point P.

4. Write two congruent statements using the diagonals of $BRHD$.

   $\overline{BH} \cong \overline{RD}$
   $\overline{BP} \cong \overline{HP}$
   $\overline{DP} \cong \overline{RP}$
   $\overline{BP} \cong \overline{HP} \cong \overline{DP} \cong \overline{RP}$



Commented [DB1]: https://www.geogebra.org/calculator/zf6n6tqk

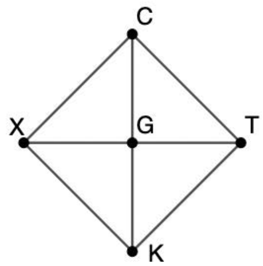
5. Given $BP = 32y - 10$, and $HP = 18y + 32$. What is the value of y?

$y = 3$

6. What is the length of $\overline{RD}$?

$RD = 172$

Quadrilateral $CTKX$ is shown with diagonals $\overline{CK}$ and $\overline{XT}$ intersecting at point G.



7. What value of g makes $CTKX$ a rhombus, if $CT = 55$, and $TK = 15g + 19$?

$g = 2.4$

8. Given that $CTKX$ is a rhombus with $CG = 6y - 17.81$ and $KG = 3y + 10.54$. What is the length of $\overline{TX}$?

$y = 9.45$
$TX = 77.78$

9. What properties are the same between $GKNV$ and $CTKX$?

Opposite sides are congruent
Opposite angles are congruent
Consecutive angles are supplementary
Opposite sides are parallel
Diagonals bisect each other

10. What properties are different between $BRHD$ and $CTKX$?

Diagonals are congruent to each other in $BRHD$
Quadrilateral $BRHD$ has four right angles